

Continuum Spectral Convergence and Time–Resolution Windows for Gaussian Quadratic Markov Operators

Collectively Compact Nyström Limits and Topology-Dependent Sampling

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

Finite Gaussian transition matrices for logarithmically driven quadratic maps have an exact parameter-response calculus, but a fixed-dimensional theorem does not by itself justify simultaneous growth of observation time and spatial resolution. We construct the corresponding fixed-noise continuum operator and derive sharp distinctions among deterministic probability propagation, weak observable estimation, and strong matrix estimation.

For $f_u(x) = 1 - ux^2$ on $[-1, 1]$ with fixed Gaussian width $\sigma > 0$, the normalized kernel defines a compact, strongly positive, operator-norm analytic Markov family $\mathcal{K}_{\sigma,u}$ on $C([-1, 1])$. Its Perron eigenvalue is simple, and all other spectral values lie strictly inside the unit disk. The midpoint Nyström operators converge strongly and collectively compactly, together with their first two parameter derivatives. Hence every nonzero isolated resonance is spectrally exact. For a simple resonance, its eigenvalue, right Nyström eigenfunction, and parameter and eigenphase derivatives converge at order h^2 , where $h = 2/d$.

Let the response scale be $s_T = (\log T)^{-p}$. A deterministic resonance response survives the joint limit if $h^2 = o(s_T)$. For row-normalized expected flow, the same result holds uniformly after one Gaussian smoothing step; if an atomic initial row is retained, the additional initial-layer condition is $d/T = o(s_T)$. Sampling introduces topology-dependent upper restrictions. With $N_i \asymp T/d$ transitions per source row, one smooth observable has error $O_{\mathbb{P}}(\sqrt{d/T})$, whereas an entire empirical row in total variation has mean error $O(d/\sqrt{T})$. Thus the leading dimension windows are

$$(\log T)^{p/2} \ll d(T) \ll \frac{T}{(\log T)^{2p}} \quad \text{for one weak observable,}$$

and

$$(\log T)^{p/2} \ll d(T) \ll \frac{\sqrt{T}}{(\log T)^p} \quad \text{for strong matrix control.}$$

Only the latter directly provides topology-robust perturbation control for a nonnormal spectrum.

Numerically, fixed-width resonance and eigenphase-response errors follow d^{-2} through $d = 50,000$. Independent row-sampling experiments recover dimension exponents 1.0046 for total variation and 0.4827 for a smooth observable, in agreement with the theory. The results neither identify a Markov resonance with a Riemann zero nor construct a self-adjoint spectral realization.

Keywords: compact Markov operator; Nyström method; collectively compact approximation; nonnormal resonance; double limit; multinomial sampling; quadratic map.

MSC 2020: 47A58; 45C05; 65R20; 37H20; 60J10.

1 Introduction

Finite transition matrices are useful numerical models only after their resolution limit is specified. This is particularly important for nonnormal Markov matrices: entrywise convergence, a visually stable phase plot, and convergence on one observable are weaker than operator control of the spectrum.

The prime-dynamics program began with a deterministic-chaos model of sieve words [10] and subsequently separated exact symbolic, Fourier, and parity statements from conjectural spectral interpretations [13]. A related numerical manuscript constructs Gaussian-smoothed transition matrices for a logarithmically driven quadratic map and compares selected eigenphases with zero ordinates [14]. A theoretical audit showed that raw phase fits can be driven by symmetry and smooth counting laws and proved fixed-resolution operator freezing [12].

The next layer derived exact first and second derivatives of the finite Gaussian matrix, proved a twice-the-tail truncation identity, and resolved the difference between an unweighted matrix average and row-normalized accumulated flow [11]. At fixed dimension, every positive-density row weighting preserves the leading unanchored response

$$(\log T)^p(Q_T - K_d(u_c)) \longrightarrow \kappa K'_d(u_c).$$

Sparse computations through dimension 50,000 exhibited second-order grid convergence of one nonreal resonance and its phase derivative.

Two questions remain. First, does the matrix spectrum converge to the spectrum of a specified continuum operator, without spectral pollution? Second, how fast may $d = d(T)$ grow while a logarithmic response remains larger than discretization and estimation errors?

The answer to the second question depends on what is estimated. Propagating the complete probability vector is deterministic and has no Monte Carlo error. Estimating one conditional expectation from a trajectory uses about T/d visits to a source bin and therefore fluctuates at scale $\sqrt{d/T}$. Estimating its entire destination distribution is harder: there are order d resolved destination cells at fixed physical noise width, and the row total-variation error is order $d/\sqrt{T}$. The two errors lead to different admissible windows.

Main results

- R1.** The fixed-width continuum Gaussian kernel defines a compact, strongly positive, analytic Markov family on $C([-1, 1])$. The eigenvalue 1 is simple and is the only peripheral spectral value.
- R2.** Midpoint Nyström operators and their first two parameter derivatives converge strongly and collectively compactly. Nonzero isolated spectral clusters have the correct algebraic multiplicity and no spurious nonzero limit.
- R3.** A simple resonance, its right Nyström eigenfunction, and its parameter and eigenphase responses converge at order d^{-2} .
- R4.** The deterministic logarithmic resonance response has a joint limit whenever $d^{-2} = o((\log T)^{-p})$. Full expected-flow normalization has the same limit after Gaussian smoothing; an included atomic initial layer contributes the additional bias $O(d/T)$.
- R5.** Conditional estimation of a single bounded observable has root-mean-square error $O(\sqrt{d/T})$. Estimation of a complete row in ℓ^1 has mean error $O(d/\sqrt{T})$, and this order persists for the maximum of independently stratified rows.
- R6.** Combining bias and variance gives distinct weak-observable and strong-matrix windows. The strong window is the appropriate sufficient condition for black-box perturbation of a nonnormal spectrum.

Scope

The continuum object in this paper is a compact Markov operator at fixed physical noise width. Its complex resonances are decay and oscillation modes, not self-adjoint energies. The six-sigma numerical width inherited from the preceding paper is historically target-informed; it is used only to audit resolution convergence and is not an arithmetic prediction. No zeta zero is loaded or compared in the new experiments, and no assertion about the Riemann hypothesis is made.

2 The fixed-noise continuum operator

Let $I = [-1, 1]$, fix $\sigma > 0$, and let u range over a compact real interval J . Put

$$f_u(x) = 1 - ux^2, \quad (1)$$

$$w_u(x, y) = \exp\left(-\frac{(y - f_u(x))^2}{2\sigma^2}\right), \quad (2)$$

$$Z_u(x) = \int_I w_u(x, z) dz, \quad q_u(x, y) = \frac{w_u(x, y)}{Z_u(x)}. \quad (3)$$

Define on $C(I)$

$$(\mathcal{K}_u g)(x) = \int_I q_u(x, y) g(y) dy. \quad (4)$$

We suppress σ when it is fixed.

Theorem 2.1 (Compact analytic Markov family). *The family (4) has the following properties.*

- (i) $\mathcal{K}_u$ is a positive contraction on $C(I)$, $\mathcal{K}_u \mathbf{1} = \mathbf{1}$, and $\mathcal{K}_u$ is compact.
- (ii) $\mathcal{K}_u$ is strongly positive: if $g \geq 0$ and $g \not\equiv 0$, then $\mathcal{K}_u g > 0$ everywhere.
- (iii) The map $u \mapsto \mathcal{K}_u$ is real analytic in operator norm on a neighborhood of J .
- (iv) Its spectral radius is one, the eigenvalue 1 is algebraically simple, and every $\lambda \in \text{spec}(\mathcal{K}_u) \setminus \{1\}$ satisfies $|\lambda| < 1$.

Proof. The normalized density is continuous and positive on the compact set $J \times I \times I$. It integrates to one in y , so positivity, contractivity, and preservation of constants follow directly. Integral operators with continuous kernels map the unit ball to a bounded equicontinuous family and are compact by Arzelà–Ascoli [7].

Strict positivity of q_u gives strong positivity. Since $Z_u(x)$ is uniformly separated from zero and the Gaussian is entire in u , all parameter derivatives of q_u are uniformly bounded on compact neighborhoods of J . Taylor expansion under the integral proves operator-norm analyticity.

Finally, $\mathcal{K}_u \mathbf{1} = \mathbf{1}$ and $\|\mathcal{K}_u\| = 1$, so the spectral radius is one. The strong form of the Krein–Rutman theorem gives simplicity of the Perron root and excludes every other peripheral eigenvalue [9]. $\square$

2.1 Continuum score derivatives

Define the score and its row mean by

$$a_u(x, y) = -\frac{x^2}{\sigma^2}(y - f_u(x)), \quad \bar{a}_u(x) = \int_I q_u(x, y) a_u(x, y) dy. \quad (5)$$

Proposition 2.2 (Exact continuum derivatives). *For $g \in C(I)$,*

$$(\mathcal{K}'_u g)(x) = \int_I q_u(x, y) (a_u(x, y) - \bar{a}_u(x)) g(y) dy, \quad (6)$$

$$(\mathcal{K}''_u g)(x) = \int_I q_u(x, y) \left[(a_u(x, y) - \bar{a}_u(x))^2 - \text{Var}_{q_u(x, \cdot)}(a_u(x, \cdot)) \right] g(y) dy. \quad (7)$$

In particular, $\mathcal{K}'_u \mathbf{1} = \mathcal{K}''_u \mathbf{1} = 0$.

Proof. Logarithmic differentiation of $q_u = w_u/Z_u$ gives the centered score in (6). The derivative $\partial_u a_u(x, y) = -x^4/\sigma^2$ is independent of y , so it cancels after centering. A second differentiation gives (7). The zero identities follow by integration. $\square$

Let $\lambda(u) \neq 0$ be a simple isolated eigenvalue, with right eigenfunction r_u and left eigenfunctional ℓ_u , normalized by $\ell_u(r_u) = 1$. Analytic compact-operator perturbation theory gives

$$\lambda'(u) = \ell_u(\mathcal{K}'_u r_u), \quad \theta'(u) = \text{Im} \frac{\ell_u(\mathcal{K}'_u r_u)}{\lambda(u)}, \quad \theta = \text{Arg} \lambda. \quad (8)$$

Because $\lambda \neq 0$, both $r_u = \lambda^{-1} \mathcal{K}_u r_u$ and the density representing ℓ_u are smooth.

3 Collectively compact Nyström approximation

Let $h = 2/d$ and

$$c_j = -1 + \left(j - \frac{1}{2}\right) h, \quad 1 \leq j \leq d.$$

Define

$$Z_{d,u}(x) = h \sum_{j=1}^d w_u(x, c_j), \quad (9)$$

$$(\mathcal{K}_{d,u} g)(x) = \frac{h \sum_{j=1}^d w_u(x, c_j) g(c_j)}{Z_{d,u}(x)}. \quad (10)$$

This is a finite-rank operator on $C(I)$. At $x = c_i$ its matrix is exactly

$$K_{ij}^{(d)}(u) = \frac{w_u(c_i, c_j)}{\sum_{k=1}^d w_u(c_i, c_k)}. \quad (11)$$

The nonzero spectra of $\mathcal{K}_{d,u}$ and $K^{(d)}(u)$ coincide, including algebraic multiplicity. Evaluation at the centers maps every nonzero finite-rank eigenfunction to a matrix eigenvector, and the Nyström formula reconstructs the eigenfunction from that vector.

Lemma 3.1 (Uniform stability). *For $u \in J$ and all sufficiently large d , $Z_{d,u}$ is uniformly bounded above and away from zero. Moreover, for $m = 0, 1, 2$,*

$$\sup_{d \geq d_0} \sup_{u \in J} \|\partial_u^m \mathcal{K}_{d,u}\|_{C(I) \rightarrow C(I)} < \infty.$$

Proof. The midpoint sums for Z_u converge uniformly in (u, x) , while $\min_{J \times I} Z_u(x) > 0$. Parameter differentiation gives finite centered score moments, bounded uniformly because I and J are compact and σ is fixed. $\square$

Theorem 3.2 (Strong and collective compact convergence). *For $m = 0, 1, 2$ and every fixed $g \in C(I)$,*

$$\sup_{u \in J} \|\partial_u^m \mathcal{K}_{d,u} g - \partial_u^m \mathcal{K}_u g\|_\infty \longrightarrow 0. \quad (12)$$

For each m , the family

$$\{\partial_u^m \mathcal{K}_{d,u} : d \geq d_0, u \in J\}$$

is collectively compact.

Proof. For fixed g , the numerator and denominator in (10), and their first two parameter derivatives, are uniform Riemann sums of continuous functions on a compact set. This proves (12).

For $\|g\|_\infty \leq 1$, differentiating (10) with respect to x gives a uniform Lipschitz bound because σ is fixed and the normalizers are separated from zero. The same applies after one or two parameter derivatives. Thus the union of the images of the unit ball is bounded and equicontinuous. Arzelà–Ascoli gives collective compactness [1, 3]. $\square$

Remark 3.3 (Why operator-norm convergence is not asserted). Midpoint evaluation cannot converge in operator norm on the whole unit ball of $C(I)$, whose elements need not share a modulus of continuity. Collective compactness plus strong convergence is the correct framework: it gives spectral exactness for nonzero isolated eigenvalues without making a false uniform quadrature claim [1, 4].

Theorem 3.4 (Spectral exactness and second-order simple branches). *Fix $u \in J$.*

- (i) *If $\lambda \neq 0$ is an isolated eigenvalue of $\mathcal{K}_u$ of algebraic multiplicity m , then every sufficiently small isolating disk contains exactly m eigenvalues of $\mathcal{K}_{d,u}$, counted algebraically, for all sufficiently large d . Every nonzero limit point of Nyström eigenvalues belongs to $\text{spec}(\mathcal{K}_u)$.*
- (ii) *If λ is simple, one can choose the Nyström branch λ_d and its continuous finite-rank right eigenfunction r_d so that*

$$|\lambda_d - \lambda| + \|r_d - r\|_\infty + |\lambda'_d - \lambda'| = O(h^2). \quad (13)$$

If $\lambda \neq 0$, then also

$$|\theta'_d - \theta'| = O(h^2). \quad (14)$$

The estimates are locally uniform along a compact simple branch.

Proof. Part (i) is the collectively compact spectral approximation theorem [1, 2, 4].

For a nonzero eigenvalue, the right eigenfunction and the density of the left eigenfunctional are smooth because they lie in the ranges of $\mathcal{K}_u$ and its adjoint. Composite midpoint quadrature is second order on these smooth functions. Standard Nyström eigenvalue estimates therefore give $|\lambda_d - \lambda| + \|r_d - r\|_\infty = O(h^2)$ [2, 3]. Apply the same argument to $\mathcal{K}'_u$ and the left-right formula (8); collective spectral projections remain uniformly bounded along an isolated simple branch. This gives the derivative estimate. The phase estimate follows from $d \text{Arg } z = \text{Im}(dz/z)$ away from zero. $\square$

Remark 3.5 (Hard Gaussian cutoff). **Theorem 3.4** concerns the full Nyström sum. If each row is truncated and renormalized, its matrix infinity-norm error is exactly twice the omitted mass [11]. A fixed six-sigma cutoff therefore approximates a fixed truncated operator to second order, while its difference from the full operator remains a small but nonzero tail perturbation. Nonnormal spectral conversion additionally depends on the left-right condition number.

4 Deterministic time-resolution limits

Let

$$u_n = u_c + \delta_n, \quad \delta_n = \frac{\kappa}{(\log(n+c))^p} + O((\log(n+c))^{-p-1}), \quad (15)$$

where $c > 1$ and $p > 0$. Put $s_T = (\log T)^{-p}$ and define the deterministic Nyström average

$$\mathcal{A}_{d,T} = \frac{1}{T} \sum_{n=1}^T \mathcal{K}_{d,u_n}. \quad (16)$$

Theorem 4.1 (Deterministic spectral double limit). *Let $\lambda(u_c) \neq 0$ be a simple isolated eigenvalue of $\mathcal{K}_{u_c}$. Let $\lambda_{d,T}$ be the corresponding eigenvalue of (16). If*

$$d(T) \longrightarrow \infty, \quad d(T)^{-2} = o(s_T), \quad (17)$$

then

$$s_T^{-1}(\lambda_{d(T),T} - \lambda(u_c)) \longrightarrow \kappa \lambda'(u_c), \quad (18)$$

$$s_T^{-1}(\text{Arg } \lambda_{d(T),T} - \text{Arg } \lambda(u_c)) \longrightarrow \kappa \theta'(u_c). \quad (19)$$

Proof. Lemma 3.1 and Taylor expansion give, uniformly in d ,

$$\mathcal{A}_{d,T} = \mathcal{K}_{d,u_c} + \kappa s_T \mathcal{K}'_{d,u_c} + o(s_T).$$

The isolated simple branch and its condition number are uniform for large d by theorem 3.4. Hence

$$\lambda_{d,T} = \lambda_d(u_c) + \kappa s_T \lambda'_d(u_c) + o(s_T).$$

Now $\lambda_d(u_c) - \lambda(u_c) = O(d^{-2}) = o(s_T)$ and $\lambda'_d(u_c) \rightarrow \lambda'(u_c)$. This proves (18); the phase formula follows similarly. $\square$

For $s_T = (\log T)^{-p}$, condition (17) is

$$d(T) \gg (\log T)^{p/2}. \quad (20)$$

There is no sampling upper bound for the unweighted deterministic average.

4.1 Row-normalized expected flow

Let $\nu_{n,d}$ be a row probability vector propagated by the full matrices,

$$\nu_{n+1,d} = \nu_{n,d} K_d(u_n),$$

and define

$$Q_{d,T;i\bullet} = \frac{\sum_{n=1}^T \nu_{n,d;i} K_{d;i\bullet}(u_n)}{\sum_{n=1}^T \nu_{n,d;i}}. \quad (21)$$

Lemma 4.2 (Uniform cell-scale occupation). *There are constants $0 < c_0 < C_0 < \infty$, independent of sufficiently large d , such that every full Gaussian matrix with $u \in J$ satisfies*

$$c_0 h \leq K_{ij}^{(d)}(u) \leq C_0 h. \quad (22)$$

Consequently, after one transition,

$$c_0 h \leq \nu_{n,d;i} \leq C_0 h \quad (23)$$

for every source row.

Proof. On the compact set $J \times I \times I$, the unnormalized Gaussian weight is between a positive constant m_σ and one. Its row sum lies between dm_σ and d . Since $d = 2/h$, (22) follows. A convex combination of the entries in a destination column has the same bounds. $\square$

Corollary 4.3 (Expected-flow double limit). *Under the hypotheses of theorem 4.1, suppose either that the initial source already satisfies (23), or that the first transition is discarded as a smoothing layer. Then the eigenvalue and phase of $Q_{d(T),T}$ satisfy (18)–(19).*

If an atomic initial row is retained, the same conclusion holds under the additional condition

$$\frac{d(T)}{T} = o(s_T). \quad (24)$$

Proof. After smoothing, all row weights are between constant multiples of h . The weighted slow-variation proof from Wang [11] is therefore uniform in d : the common factor h cancels between numerator and denominator. Uniform parameter derivatives then give

$$Q_{d,T} = K_d(u_c) + \kappa s_T K'_d(u_c) + o(s_T)$$

in matrix infinity norm. Apply the proof of theorem 4.1.

An atomic source contributes at most one unsmoothed weight to one row. Its row denominator is of order Th , so the resulting initial-layer term is $O((Th)^{-1}) = O(d/T)$. Condition (24) makes it $o(s_T)$. $\square$

Remark 4.4 (Sparse cutoff and thresholds). The uniform lower bound (22) uses the full Gaussian kernel. A hard cutoff or a numerical rule that discards source masses below a threshold introduces exact zeros and can invalidate uniform occupation. Those modifications require a separate reachability and denominator estimate.

5 Sampling topology and empirical windows

The deterministic flow (21) propagates all probabilities; it is not a one-trajectory transition-count estimator. We now quantify the additional cost of empirical estimation.

For row i , suppose N_i destination samples $Y_{i,1}, \dots, Y_{i,N_i}$ are independent, with possibly nonidentical probabilities $p_{i,k} \in \mathbb{R}^d$. Let

$$\bar{p}_i = \frac{1}{N_i} \sum_{k=1}^{N_i} p_{i,k}, \quad \hat{p}_i = \frac{1}{N_i} \sum_{k=1}^{N_i} e_{Y_{i,k}}.$$

This row-stratified model isolates conditional transition sampling. It also applies conditionally in settings where visits to each source row supply an appropriate concentration inequality.

Theorem 5.1 (Strong and weak row errors). *Assume $N_i \asymp T/d$ uniformly in i .*

(i) *For every row,*

$$\mathbb{E} \|\hat{p}_i - \bar{p}_i\|_1 \leq \sqrt{\frac{d}{N_i}} = O\left(\frac{d}{\sqrt{T}}\right). \quad (25)$$

If rows are sampled independently, then

$$\mathbb{E} \max_{1 \leq i \leq d} \|\hat{p}_i - \bar{p}_i\|_1 = O\left(\frac{d}{\sqrt{T}} + \sqrt{\frac{d \log d}{T}}\right) = O\left(\frac{d}{\sqrt{T}}\right). \quad (26)$$

(ii) For every $g \in \mathbb{R}^d$ with $\|g\|_\infty \leq 1$,

$$\left(\mathbb{E}|(\hat{p}_i - \bar{p}_i)g|^2\right)^{1/2} \leq \frac{1}{\sqrt{N_i}} = O\left(\sqrt{\frac{d}{T}}\right). \quad (27)$$

For a fixed finite collection of rows and observables, the same order holds jointly. Uniform control over all d rows adds at most a $\sqrt{\log d}$ factor.

Proof. For coordinate j , independence and Jensen's inequality give

$$\mathbb{E}|\hat{p}_{ij} - \bar{p}_{ij}| \leq \sqrt{\frac{\bar{p}_{ij}}{N_i}}.$$

Summing and applying Cauchy–Schwarz proves (25). Changing one destination sample changes the row ℓ^1 error by at most $2/N_i$. The bounded-differences inequality [8], followed by a union bound over rows, adds $O(\sqrt{\log d/N_i})$ above the largest mean. This proves (26).

For (27), the centered summands $g(Y_{i,k}) - \mathbb{E}g(Y_{i,k})$ have absolute value at most two and variance at most one. Variances add and division by N_i^2 gives the claim. $\square$

Remark 5.2 (Sharper Gaussian-row mean bound). For identically distributed multinomial samples with row probability p_i ,

$$\mathbb{E}\|\hat{p}_i - p_i\|_1 \leq \frac{1}{\sqrt{N_i}} \sum_j \sqrt{p_{ij}(1 - p_{ij})}.$$

At fixed physical σ , $p_{ij} \asymp h$ over order d resolved cells, so the right side is a constant multiple of $d/\sqrt{T}$. The numerical experiment below uses this bound.

5.1 Three response windows

The deterministic spectral bias is $O(d^{-2})$. Combining it with [theorem 5.1](#) gives the following sufficient regimes.

Corollary 5.3 (Topology-dependent logarithmic windows). *Let $s_T = (\log T)^{-p}$.*

(a) *Deterministic unweighted propagation requires only*

$$(\log T)^{p/2} \ll d(T). \quad (28)$$

Row-normalized deterministic flow from an included atomic source also requires $d(T) \ll T/(\log T)^p$.

(b) *A fixed finite collection of weak observables is controlled when*

$$(\log T)^{p/2} \ll d(T) \ll \frac{T}{(\log T)^{2p}}. \quad (29)$$

Uniform weak control over all rows replaces the upper condition by $d(T) \log d(T) \ll T/(\log T)^{2p}$.

(c) *Strong row-total-variation control, and hence robust matrix infinity-norm perturbation of a uniformly conditioned isolated branch, holds when*

$$(\log T)^{p/2} \ll d(T) \ll \frac{\sqrt{T}}{(\log T)^p}. \quad (30)$$

Proof. The lower bound is $d^{-2} = o(s_T)$. The weak upper bound is $\sqrt{d/T} = o(s_T)$, and the strong upper bound is $d/\sqrt{T} = o(s_T)$. The atomic-source condition follows from (24). $\square$

Table 1: Errors relative to the fitted leading h^2 extrapolation.

d	$ \lambda_d - \lambda_\infty^{(6\sigma)} $	$ \theta'_d - (\theta')_\infty^{(6\sigma)} $
2,000	2.3574×10^{-4}	2.0381×10^{-2}
5,000	3.7518×10^{-5}	3.2455×10^{-3}
10,000	9.3746×10^{-6}	8.1098×10^{-4}
20,000	2.3455×10^{-6}	2.0290×10^{-4}
50,000	3.7790×10^{-7}	3.2674×10^{-5}

Remark 5.4 (Weak convergence does not control a nonnormal spectrum). Agreement on one or several smooth observables can hold far outside the strong window. It does not bound the matrix infinity norm or the resolvent of a nonnormal operator. A weaker spectral theorem is possible only after using additional regularity of the particular left and right eigenfunctions; it is not supplied by an observable RMS plot alone.

Remark 5.5 (One trajectory). [Theorem 5.1](#) is a row-stratified theorem. A single non-autonomous Markov trajectory also requires uniform source-count and conditional concentration estimates as d grows. At fixed d , Gaussian positivity suffices [\[11\]](#); uniform $d(T)$ control is a mixing problem and is not inferred here from pointwise positivity.

For an endpoint-anchored response of order $(\log T)^{-p-1}$, every occurrence of p in [corollary 5.3](#) must be replaced by $p + 1$. Anchoring therefore makes the empirical window narrower even though it cancels the leading deterministic drift.

6 Numerical verification

6.1 Nyström resonance extrapolation

We reuse the machine-readable fixed-width benchmark from Wang [\[11\]](#). Its parameters are

$$u_c = 1.5437, \quad \sigma = 0.00785, \quad L = 6,$$

and the selected branch is the largest-modulus nonreal resonance in the open upper half-plane. The width was inherited from historically target-informed exploratory scans; no zeta-zero data enter the present extrapolation.

For $d = 2000, 5000, 10000, 20000, 50000$, fit the real part, imaginary part, and phase derivative to the predicted leading form

$$b_0 + b_2 h^2$$

using $d \geq 5000$. The extrapolated six-sigma values are

$$\lambda_\infty^{(6\sigma)} = -0.473827327363 + 0.506120759277 i, \tag{31}$$

$$(\theta')_\infty^{(6\sigma)} = 2.282035521438. \tag{32}$$

The superscript records that this is a truncated-kernel extrapolation, not an exact full-Gaussian value.

The agreement in [figure 1](#) verifies the rate predicted by [theorem 3.4](#) for this isolated branch. It does not by itself prove the continuum theorem, and the fitted limit is not an arithmetic target.

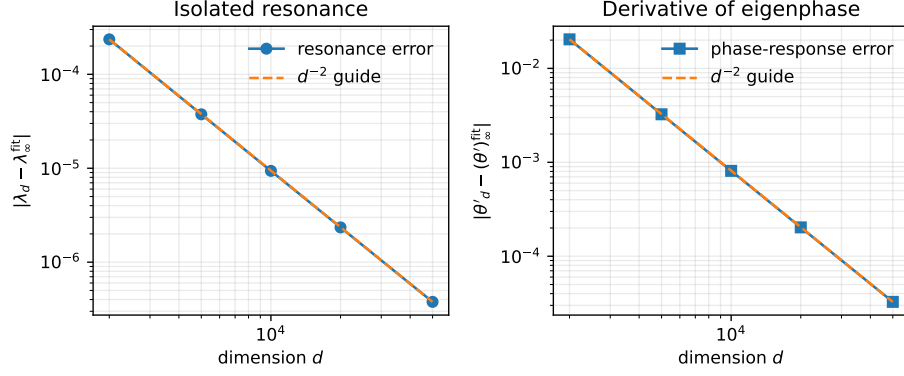


Figure 1: Second-order Nyström convergence to the fitted six-sigma continuum branch. Both the nonreal resonance and its analytic eigenphase derivative follow the d^{-2} guide across the full measured range.

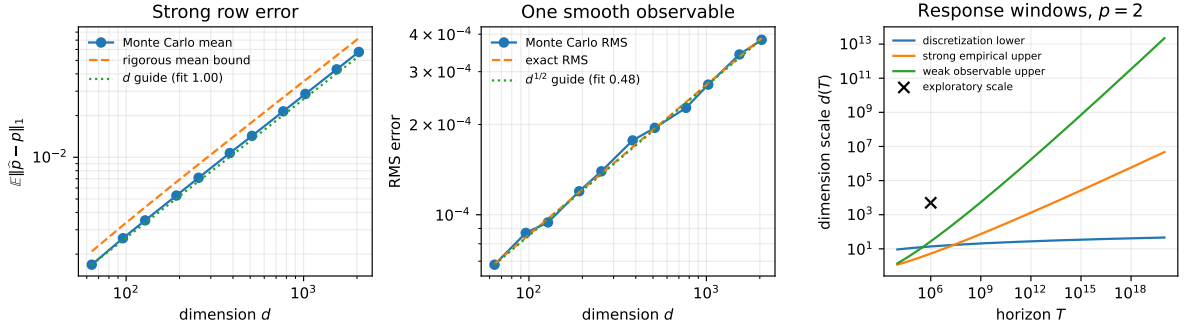


Figure 2: Sampling topology and response windows. Left: complete-row total-variation error follows $d/\sqrt{T}$ and stays below the rigorous multinomial mean bound. Middle: one smooth observable follows $\sqrt{d/T}$ and agrees with its exact multinomial RMS. Right: the deterministic lower scale and the weak and strong upper scales for $p = 2$. The marked exploratory point used deterministic probability propagation and therefore is not subject to the empirical upper curves.

6.2 Strong versus weak conditional sampling

To isolate sampling topology, choose a representative Gaussian row at $u = 1.5437$, source position $x = 0.8$, and the untuned width $\sigma = 0.05$. For dimensions from 64 through 2048, assign

$$N_d = \left\lfloor \frac{10^8}{d} \right\rfloor$$

independent destination samples and repeat the multinomial experiment 500 times. The weak observable is

$$g(y) = \cos(\pi y) + 0.3 \sin(2\pi y).$$

The scripts use NumPy for sampling and Matplotlib for the publication figures [5, 6].

The fitted dimension exponents over $d \geq 256$ are

$$\mathbb{E} \|\hat{p} - p\|_1 \asymp d^{1.0046}, \quad \left(\mathbb{E} |(\hat{p} - p)g|^2 \right)^{1/2} \asymp d^{0.4827}, \quad (33)$$

at fixed total transition budget. They agree with the theoretical exponents 1 and 1/2.

For $p = 2$ and $T = 10^6$, the unit-constant scales are

$$(\log T)^{p/2} = 13.82, \quad \frac{\sqrt{T}}{(\log T)^p} = 5.24, \quad \frac{T}{(\log T)^{2p}} = 27.45.$$

Thus there is not yet a unit-constant strong empirical window, while a narrow weak window is visible. These values are illustrative, because asymptotic little- o conditions do not specify universal finite constants. More importantly, a deterministic full-cloud computation has no sampling upper bound at all. It can still retain an atomic initial-layer bias: at $d = 5000$ and $T = 10^6$,

$$\frac{d/T}{(\log T)^{-2}} \approx 0.95,$$

so the first unsmoothed source row is not negligible on the nominal response scale unless it is discarded or separately subtracted.

7 Implications for quadratic spectral models

7.1 What the continuum theorem supplies

At fixed physical noise, the finite matrices are not free-standing spectral lists. They are Nyström representations of the explicitly defined compact operator (4). Nonzero isolated matrix resonances converge with multiplicity, and a simple branch has a controlled derivative. This turns the observed d^{-2} law into a consequence of smooth quadrature rather than an unexplained numerical pattern.

The theorem also identifies what must be centered. The unscaled long exposure freezes to $\mathcal{K}_{u_c}$; the first logarithmic spectral response is the derivative (8). Grid refinement can accompany this limit only when the deterministic branch bias is $o(s_T)$.

7.2 Why the estimator must be named

Three objects that look similar in code have different admissible limits:

- (i) an unweighted average of deterministic kernels has only the discretization lower bound;
- (ii) row-normalized expected flow also needs control of any atomic initial layer but has no Monte Carlo variance; and
- (iii) a single-trajectory conditional matrix must resolve both source visits and destination distributions.

Calling all three a transition matrix hides the factor $\sqrt{d}$ separating weak and strong estimation.

7.3 What remains open

The present result fixes $\sigma > 0$. If σ shrinks with h , the continuum operator changes toward deterministic composition and the raw matrix derivative norms can diverge [11]. A triple limit in T, d, σ therefore requires a different Banach topology.

Likewise, operator averages forget temporal order. After continuum and sampling control, the next order-sensitive object should be a parity-resolved two-step cocycle or a centered trace response [13]. Any comparison with arithmetic fluctuations would then require target-independent parameters, independent unfolding, and a separate theorem. None is asserted here.

8 Conclusion

The fixed-noise Gaussian quadratic model has a rigorous continuum spectral meaning. Its normalized kernel is a compact, strongly positive analytic Markov operator, and midpoint Nyström matrices are collectively compact spectral approximations. Every nonzero isolated resonance is free of nonzero spectral pollution; simple resonance and eigenphase responses converge at second order.

The joint time–resolution problem is not described by one universal window. Deterministic probability propagation needs only to make the d^{-2} bias smaller than the logarithmic response. A weak observable estimated from conditional samples costs $\sqrt{d/T}$, while a complete row in total variation costs $d/\sqrt{T}$. Strong matrix control is consequently much more restrictive than a successful observable plot, exactly as the multinomial experiments demonstrate.

These results complete the continuum and sampling foundation for the renormalized Gaussian response. They do not convert Markov resonances into self-adjoint energies, identify them with Riemann zeros, or imply the Riemann hypothesis.

Data and code availability

The manuscript source, tests, machine-readable results, and figure scripts are available at https://github.com/maris205/prime_dynamics_theory/tree/main/papers/continuum-spectral-double-limits. The spectral extrapolation reads the published benchmark data from the preceding paper in the same repository.

Acknowledgments

The author acknowledges the use of an AI language model for assistance with mathematical cross-checking, code development, manuscript organization, and typesetting. The author verified the final arguments and assumes responsibility for the content.

References

- [1] Philip M. Anselone. *Collectively Compact Operator Approximation Theory and Applications to Integral Equations*. Prentice-Hall, Englewood Cliffs, NJ, 1971.
- [2] Kendall E. Atkinson. The numerical solution of the eigenvalue problem for compact integral operators. *Transactions of the American Mathematical Society*, 129(3):458–465, 1967. doi: 10.1090/S0002-9947-1967-0223112-0.
- [3] Kendall E. Atkinson. *The Numerical Solution of Integral Equations of the Second Kind*. Cambridge University Press, Cambridge, 1997. doi: 10.1017/CBO9780511626340.
- [4] Françoise Chatelin. *Spectral Approximation of Linear Operators*. Academic Press, New York, 1983.
- [5] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- [6] John D. Hunter. Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi: 10.1109/MCSE.2007.55.
- [7] Rainer Kress. *Linear Integral Equations*, volume 82 of *Applied Mathematical Sciences*. Springer, New York, 3 edition, 2014. doi: 10.1007/978-1-4614-9593-2.
- [8] Colin McDiarmid. On the method of bounded differences. In *Surveys in Combinatorics, 1989*, volume 141 of *London Mathematical Society Lecture Note Series*, pages 148–188. Cambridge University Press, 1989. doi: 10.1017/CBO9781107359949.008.

- [9] Helmut H. Schaefer. *Banach Lattices and Positive Operators*. Springer, Berlin, 1974. doi: 10.1007/978-3-642-65970-6.
- [10] Liang Wang. The emergence of prime distribution from low-dimensional deterministic chaos. *Research in Mathematics*, 13(1):2684334, 2026. doi: 10.1080/27684830.2026.2684334.
- [11] Liang Wang. Renormalized spectral response of gaussian quadratic transfer matrices: Exact derivatives, occupation weighting, and resolution scaling. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/renormalized-gaussian-response.
- [12] Liang Wang. Spectral obstructions and renormalized limits for quadratic models of Riemann-zero statistics. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/spectral-obstructions-renormalized-limits.
- [13] Liang Wang. Parity-resolved dynamics at a quadratic band-merging parameter: Peripheral spectrum, periodograms, and two-step sequential stability. Preprint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/parity-resolved-band-merging-spectrum.
- [14] Liang Wang. Spectral isomorphism between renormalization flow in non-autonomous quadratic maps and Riemann zeros. Manuscript, 2026. URL https://github.com/maris205/riemann_logistic.